

Titanic Dataset – Exploratory Data Analysis

Introduction

This project performs Exploratory Data Analysis (EDA) on the Titanic dataset using Python, Pandas, Matplotlib, and Seaborn. The analysis focuses on understanding passenger characteristics, survival patterns, relationships between variables, missing values, and outliers.

Key Findings

- 38.4% of passengers survived, while 61.6% did not.
- Female passengers had a much higher survival rate (74.2%) than male passengers (18.9%).
- First-class passengers had the highest survival rate (63.0%), while third-class passengers had the lowest (24.2%).
- Passengers who paid higher fares generally had better survival outcomes.
- Passengers from Port C had the highest survival rate (55.4%).
- Small family groups generally showed better survival rates.
- `Cabin` had the most missing values (687), followed by `Age` (177).
- The `Fare` column contained several high-value outliers.

Conclusion

The EDA shows that **gender, passenger class, and fare were important factors associated with survival**. Female and higher-class passengers generally had better chances of survival. The analysis also identified missing values and fare outliers, providing useful insights into the Titanic dataset.